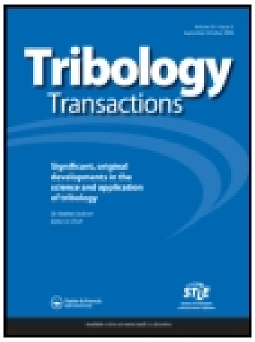

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Ball Bearing and Tapered Roller Bearing Torque: Analytical, Numerical and Experimental Results

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Ball Bearing and Tapered Roller Bearing Torque: Analytical, Numerical and Experimental Results©

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The miscellaneous components of bearing torque are described analytically.

When the rolling element - race contact lengths are similar, the hydrodynamic ball bearing torque component can be similar to the tapered roller bearing one.

In ball bearings, the sliding speed distribution encountered at the ball/race contact is the result of two torque contributions: curvature in space of the contact ellipse and spinning effects.

These two torque contributions don't exist in tapered roller bearings. However, tapered roller bearings are subjected to a

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NOMENCLATURE

a	= half length of the ball – race contact ellipse
a_1, a_2	= location of the rolling line
b	= half width of the rolling element – race contact
BB	= ball bearing
dF^s	= sliding force in a slice of the contact ellipse
D	= ball diameter
d_m	= pitch diameter
dQ	= load in a slice of the contact
dT	= final torque due to one rolling element
dTC	= one ball torque contribution due to MC
$dTER$	= one rolling element contribution due to MER
dTP	= one ball contribution due to MP
dTR	= one rolling element contribution due to FR
$dTRIB$	= one roller contribution due to FRIB and MRIB
E	= dimensionless parameter used in BB calculation
f	= roller – rib contact height, or ball – race osculation ratio R_{yi}/D and R_{yo}/D
f_c	= factor used for calculating MC
f_p	= factor used for calculating MP
FR	= hydrodynamic rolling force
$Frib$	= rib sliding force
FS	= Entrainment (or sliding) force
G	= dimensionless material parameter (used for calculating FR)
k	= ratio of the equivalent radius R_y/R_x
K	= factor used for defining the race curvature radius
L	= effective length of the roller
MC	= moment due to the curvature effect
MER	= moment due to the elastic rolling effect
MP	= moment due to the pivoting effect
$Mrib$	= moment due to the rib effect
P_{max}	= maximum Hertzian pressure
Q	= rolling element – race load
R_a	= radius of the curved Hertzian contact
R_i	= mean inner race radius
R_o	= mean outer race radius
R_x, R_y	= equivalent radius in the x and y direction
TRB	= tapered roller bearing
U	= dimensionless speed and viscosity parameter (used for calculating FR)
W	= dimensionless load parameter (used for calculating FR)
x	= rolling direction; X: radial direction
y	= direction along the rolling element – race contact.; Y: rotational axis
z	= direction normal to the contact; Z: radial direction
α	= rolling element – race contact angle
ω_c	= orbital rotational speed of the rolling element
ω_r	= rotational rolling speed of the rolling element
μ	= friction coefficient
μ_a	= asperity friction coefficient
μ_e	= equivalent friction coefficient
ν	= half included roller angle
Λ	= ratio film thickness/ RMS surface roughness height

INDEX

i	= inner race
b	= ball or rolling element
o	= outer race
lc	= line contact
pc	= point contact
rib	= rib

TABLE 1—CALCULATED HERTZIAN PARAMETERS

	TRB	BB		
		K=0.03	K=0.06	K=0.10
p_{max} (Mpa)	1200	2600	3090	3500
2b (mm)	0.392	0.870	0.991	1.089
2a (mm)	L=30	8.894	6.817	5.648

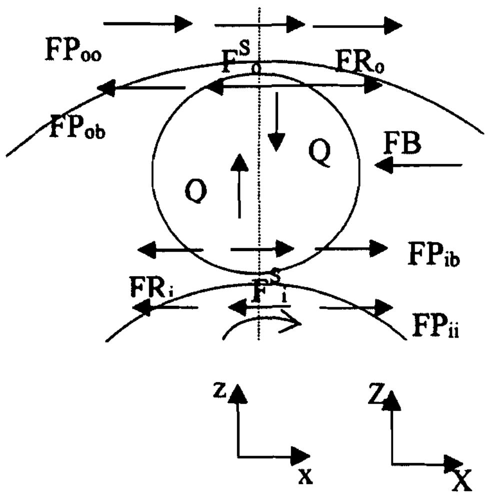

Fig. 1—List of forces.

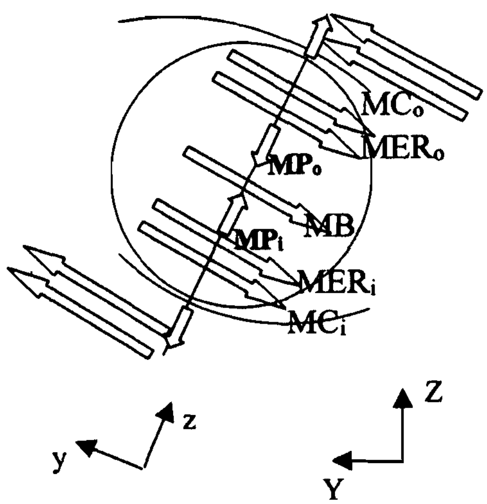

Fig. 2—List of moments.

roller/rib torque which is compared analytically to the two previous ball bearing torque components.

The final tapered roller bearing torque is compared to ball bearing torque under various operating conditions. It is shown that there are conditions for which the tapered roller bearing torque can be smaller than the ball bearing one; the latter point is demonstrated numerically and experimentally in reported literature.

KEY WORDS

Ball and Tapered Roller Bearing Torque; Power Losses; Ball Bearings, Tapered Roller Bearings

OBJECTIVES

There is growing interest in industry for power loss reduction. The objective of this paper is to explain how the bearing torque of ball bearings (BB) and tapered roller bearings (TRB) are calculated so that a comparison about bearing torque can be made and the best bearing selected as a function of the application.

Emphasis must be put on analytical relationships which facilitate the optimization of bearing design and also the understanding of the ideal operating conditions.

Finally, it is always useful to deliver to the engineering community some additional educational material on bearing torque calculations.

HERTZIAN CONTACT

One of the main differences between a TRB and BB lies in the rolling element – race contact which is described using the Hertzian line contact or point contact, respectively.

In a Hertzian line contact, the maximum Hertzian contact pressure P_{max} is uniform in the y direction (along the roller) and elliptical in the x direction (along the contact width of length 2b), whereas in a point contact, the pressure is elliptical in both direction of the contact ellipse of dimension 2b and 2a.

Relationships for calculating a, b and P_{max} are given by Houpert in (Houpert, (1996), (1999), (2000)) and require the knowledge of the race radius of curvature R_y that is described by introducing the factor K:

$$R_y = (1 + K)\frac{D}{2} \qquad [1]$$

where D is the rolling element diameter. K varies between 0.02 (for precision bearing) to 0.10, a typical value being 0.05

The Hertzian contact in a TRB is flat in the y direction while it is curved in spaced in a BB, the Hertzian radius of curvature of the elastic Hertzian contact being R_a:

$$R_a = \frac{D}{2}(\frac{1+K}{1+\frac{K}{2}}) \qquad [2]$$

It will be shown later that, due to the curvature of the contact ellipse, a substantial amount of sliding speed occurs at the ball – race contact, while almost pure rolling conditions occur in a TRB.

The contact length in a TRB is usually longer the one in a BB because the ellipse length 2a in a BB is of the order of D/2 while the roller length L in a TRB is in general of the order of 1.5 D. Consequently, at a given rolling element - race load Q, the maximum pressure P_{max} in a TRB is substantially lower than the one in a BB. (Such a lower pressure can increase significantly the TRB life, but this is not the topic of this paper).

Table 1 compares the Hertzian parameters calculated for different values of K and for a given case: Q = 11085 N, D = 20 mm and L = 30 mm.

BEARING TORQUE MODEL

Bearing torque model for TRB is described by R. Zhou and M. Hoeprich in (Zhou, et al., (1991)), while bearing torque model for BB is described by Houpert in (Houpert, (1997), (1999), Houpert, et al., (1985)). As explained in these papers, torque can be calcu-

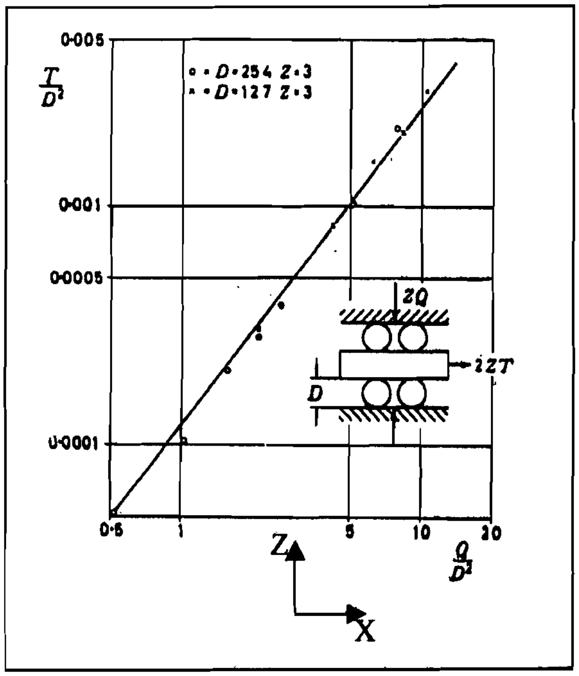

Fig. 3—The elastic rolling resistance as measured by Snare.

lated by knowing the miscellaneous entrainment force (named F^S in Fig. 1), and braking forces and moments applied on the rolling elements as shown in Figs. 1 and 2. These forces and moments must be in equilibrium and can be calculated as a function of the load, viscosity, entrainment speed (or rolling speed), sliding speed, local friction coefficient and bearing geometry.

FR are hydrodynamic rolling forces that will be described later; FP are the forces resulting from the non symmetrical pressure distribution along the contact. They can be expressed as a function of FR. MC and MP are braking moments due to the curvature of the contact and pivoting effects. These effects induce a certain amount of positive and negative sliding speed, hence local friction force at the ball race contact. Houpert, (Houpert, (1999)), shows how the sliding speed can be calculated, especially the locations of the rolling line named a_2 and a_1 in Figs. 5 and 6. How MP and MC are calculated will be described later.

Last MER is described in (Houpert, (1999)) as the braking elastic rolling moment and is calculated as a function of the load, ball diameter, and local friction coefficient.

In TRB, sliding speed is nil (or almost nil) at the roller – race contact, but slip is found at the roller – rib contact where a small load Q_{rib} is applied. As a result, a sliding force Frib and braking rib moment Mrib can be calculated and will be described later.

HYDRODYNAMIC RACE TORQUE

A common contribution in TRB and BB to the final bearing torque are the hydrodynamic losses which are calculated using hydrodynamic resistant forces FR. Relationships for calculating FR are available for point contact and line contact as described by Houpert in (Houpert, (1987)).

The hydrodynamic force used in a point contact (ball – race contact) is:

$$FR_{pc} = 2.86E.R_x^2.k^{0.348}.G^{0.022}.U^{0.66}.W_{pc}^{0.47} \quad [3]$$

where G, U and W are the classical material, speed and load parameters also used for film thickness calculation, R_x is the equivalent radius in the rolling direction, and k is the equivalent radius ratio. Note that the speed parameter U is in fact proportional to the product of viscosity times entrainment speed.

In (Houpert, (1999)) are also given relationships for calculating the contact ellipse length 2a:

$$2.a = 2.31R_x.k^{0.4037}.W_{pc}^{0.33} \quad [4]$$

It can therefore be seen that FR is proportional to the contact length 2a and slightly load dependent:

$$FR_{pc} = 1.23ER_xk^{-0.05}G^{0.022}U.^{0.66}W_{pc}^{0.14}.2.a \quad [5]$$

For line contact, it can be found in the same reference:

$$FR_{lc} = 0.04E.R_x.L.U^{0.44}.W_{lc}^{0.37} \quad [6]$$

Hence for TRB and BB, the hydrodynamic rolling forces are a function of the product (viscosity*speed), are slightly a function of the applied load and are proportional to the contact length L (for a line contact) or 2*a for a point contact.

It can be checked that, providing the contact length are similar in a BB and TRB (condition required for having similar pressure and life), the hydrodynamic forces are of the same order of magnitude in BB and TRB, especially at low to medium speed.

Note that when this condition occurs (2*a ≈ L), the BB diameter (D = 4*a=2*2a) is larger than the roller one (D = 0.66 L) inducing a larger BB outer race, hence larger BB torque as shown analytically and numerically (with an example) at the end of the paper.

At very high speeds, BB are often used because they are less sensitive to centrifugal effects.

MER MOMENT

Another common braking moment in BB and TRB is the Elastic Rolling Moment, called MER in (Houpert, (1999)).

It has been measured for ball and roller contact by Snare in (Snare, (*Rolling Resistance in Bearing*)) as shown on Fig. 3 and attributed to Hysteresis losses: rollers are pressed between three plates (slightly lubricated). A force T (transformed into a moment MER) is needed to roll the roller slowly.

$$MER = 0.00021.Q.D \quad [7]$$

It is believed by the author that the resistance to rolling motion is not only due to hysteresis losses, but more so to the rolling creep effects studied by K. L. Johnson, and described by Todd and Stevens in (Todd, et al., (*ESA Trib.*)) or Harris in (Harris, (1990)).

Figure 4 describes the stick and slip regions in the Hertzian contact.

The stick – slip phenomenon is controlled by the local friction coefficient. Slip occurs when the tangential stress (needed to tan-

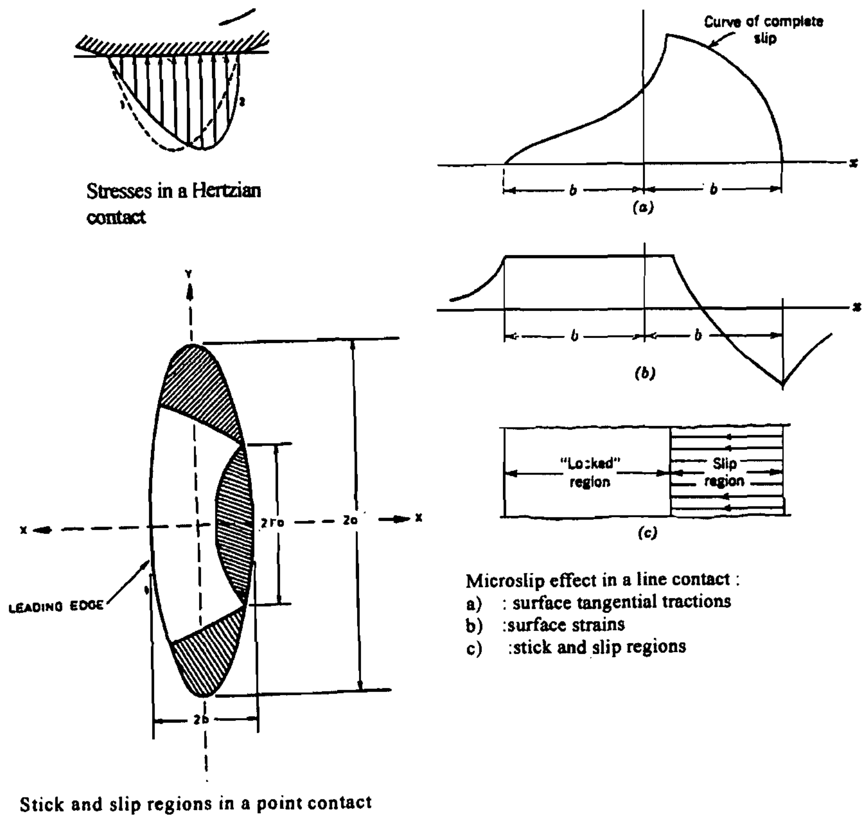

Fig. 4—The micro-slip concept in a line and point contact as described in (Harris, (1990)).

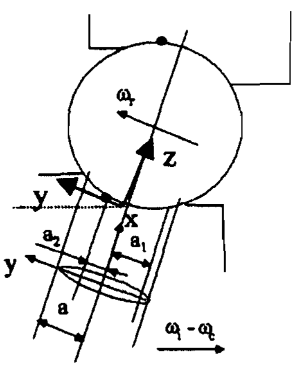

Fig. 5—Rolling line locations.

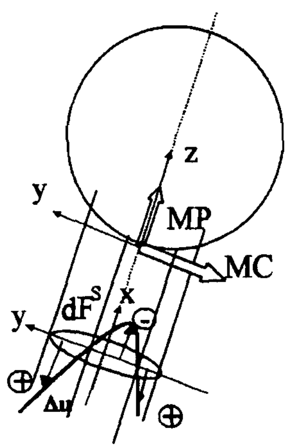

Fig. 6—Sliding speed distribution.

gentially deform the contact in the rolling direction) has reached its maximum value defined by the maximum friction coefficient.

When the maximum friction coefficient varies, the previous relationship should therefore be approximately modified using the following relationship:

$$MER = 0.00042.Q.R_x.\frac{\mu_e}{0.11} \quad [8]$$

How the friction coefficient, or its equivalent value μ_e combining lubricant shearing and asperity effects, is calculated will be described later. At low speed, it is of the order of 0.11, while at large film thickness (relative to surface roughness height), it is of the order of 0.05 in BB (because slip always occurs in BB) and zero in TRB (because almost pure rolling is found at the roller – race contact).

MC AND MP MOMENT

Figures 5 and 6 describe the sliding speed distribution occurring at the ball – race contact, as well as the resulting friction moment MC (due to curvature effects) and MP (due to pivoting effects).

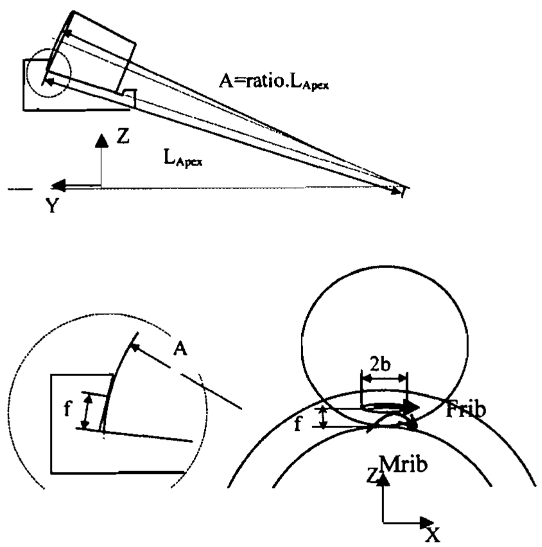

Fig. 7—Friction and moment at roller – rib contact.

The sliding speed distribution and the location of the rolling lines a_1 and a_2 have been calculated in (Houpert, (1999)) as a function of a single parameter E and the final entrainment force F^S.

$$E \approx \frac{2.R_a}{a}.\frac{D.sin\alpha}{d_m} \qquad [9]$$

where R_a is the radius of curvature of the Hertzian contact, d_m the ball pitch diameter and α the contact angle.

For E < 1, two rolling lines are found in the Hertzian contact and the sliding speed, hence friction force dF^S in a given slice, changes twice in sign between – a and + a.

For E > 1, one rolling line is found, often near the ellipse center so about half of the ellipse is under negative sliding force dF^S, the other one under positive force.

By expressing dF^S as $\mu.dQ$, with dQ being the load on a given slice and μ the friction coefficient, positive or negative and assumed constant in absolute value, the final moment MC and MP can be calculated as:

$$MC = \int_{-a}^{+a} dF^S.z \qquad MP = \int_{-a}^{+a} dF^S.y$$

$$\text{with } z \approx \frac{y^2}{2.R_a} \qquad [10]$$

or

$$MC = 0.0806\mu.Q.\frac{a^2}{R_a}.f_c \qquad [11]$$

$$MP = \frac{3}{8}.\mu.Q.a.f_p \qquad [12]$$

with the factor f_c and f_p being mainly a function of E and varying in opposite direction from 0 to 1.

Analytical relationships are given in (Houpert, (1999)) for calculating the location of the rolling line, as well as f_c and f_p, as a function of E and F^S, the final entrainment force needed to keep the rolling element rolling. F^S is usually very small and the following curve-fitted relationship applies for F^S assumed close to zero.

$$f_c \approx 1.0026 - 0.1653E - 0.2638$$

$$E^2 - 2.5521E^3 + 1.9749E^4$$

$$f_p \approx 0.0042 + 1.1045E + 0.4625E^2 - 0.5648E^3 \qquad [13]$$

Under pure radial load, the contact angle α is nil, E is nil, and symmetrical rolling lines are found at ± 0.34 a. The factor f_c is then equal to 1 and f_p to 0, i.e., MC is maximum and MP is nil.

Under a combined axial and radial load, the contact angle gets larger and E gets rapidly larger than 1. One rolling line is then found (at $y \approx 0$) and the factor f_c is nil while f_p is equal to 1, i.e., MP is maximum and MC is nil.

RIB TORQUE IN TRB

Figure 7 shows the rib friction force Frib and moment Mrib that can be calculated approximately using:

$$Frib \approx \mu_{rib}.Q_{rib}$$

$$Mrib \approx \mu_{rib}.Q_{rib}.f\sqrt{1 + 0.18(\frac{b}{f})^2} \qquad [14]$$

where μ_{rib} is the local friction coefficient and Q_{rib} the small roller – rib load calculated approximately as:

$$Q_{rib} \approx 2.Q.sin\nu \qquad [15]$$

where ν is half the included roller angle, of the order of 2 degrees. The contact height f, is calculated as a function of the bearing geometry and is often of the order of 1 mm.

FRICTION COEFFICIENT

All previous relationships, used for calculating MC, MP, MER and Mrib, require the knowledge of the friction coefficient μ.

Friction coefficient is due to the shearing of the lubricant film thickness and also asperity contact when the Λ ratio (i.e. film thickness/RMS roughness height) is small.

Lubricant friction coefficient μ_{oil} has been extensively described by Houpert (see for example (Houpert, et al., (1981), 1984)) using thermal visco - elastic – plastic rheological models. The lubricant friction coefficient increases with sliding speed (or rather lubricant shear rate) before reaching a maximum value, function of the operating pressure, temperature and lubricant type, and then decreases at large sliding speed because of temperature effects.

Maximum friction coefficient is of the order of 0.05 for mineral oil and 0.02 for synthetic oil.

Friction due to asperities has been calculated by Houpert in (Houpert, et al., (1984)) by using the Λ ratio for defining the deformation on the top of the asperities, hence pressure and load Q_a carried by the asperities. The asperity pressure and load are mainly a function of the Λ ratio, but also RMS asperity slope.

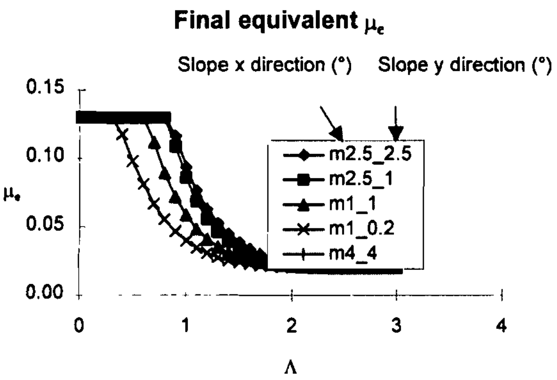

Fig. 8—Roller – rib friction coefficient, function of Λ and asperity slope.

Using a boundary friction coefficient μ_a on the asperities, a friction force due to asperity contact can be defined as:

$$F_a = \mu_a.Q_a \qquad [16]$$

By adding the lubricant and asperity friction force, the final equivalent friction coefficient can be defined:

$$F^S = \mu_a.Q_a + \mu_{oil}.(Q - Q_a) = \mu_e.Q \qquad [17]$$

$$\mu_e = \mu_a.\frac{Q_a}{Q} + \mu_{oil}(1 - \frac{Q_a}{Q}) \qquad [18]$$

Figure 8 shows an example of the calculated friction coefficient at the roller – rib contact with $\mu_a = 0.13$, $\mu_{oil} = 0.02$ and different values of asperity slopes defined in the x and y direction.

FORCE AND MOMENT EQUILIBRIUM

All the previously described forces and moments are functions of rolling and sliding speed. The forces and moments equilibrium could then be used for deriving the bearing kinematics: the orbital and rotational speed (ω_r and ω_c), for example.

But many of the previous forces and moments can be calculated using the nominal kinematics (kinematics defined using pure rolling conditions at the rolling lines of the element – race contact). Once the rolling line location at the ball – race contact is known , the sliding speed elsewhere is known, and the calculation of the moment MP and MC is possible.

The nominal kinematics is also sufficient for defining the rolling speed (or entrainment speed) needed for calculating hydrodynamic film thickness and rolling force FR, and is also sufficient for defining the roller – rib sliding speed, hence rib friction force Frib and moment Mrib.

The nominal kinematics is however not sufficient for defining the rolling element – race entrainment (or sliding) force F^S (at both contact) which are very sensitive to the sliding speed. A small variation of the ball rotational speed indeed induces a large variation of the gross sliding speed, hence sliding force. The very non linear behavior of the sliding force *vs.* sliding speed complicates therefore the calculation of ω_r and ω_c.

But by choosing the two latter forces F^S_i and F^S_o as the author's new unknowns (instead of the orbital and rotational speed), the solving of two forces and moments equilibrium linear equations easily give an analytical relationship for these two unknowns:

For BB the following relationship is shown in (Houpert, (1999)):

$$F_i^S = \frac{MC_o + MC_i + MER_i + MER_o + MB}{D} + FR_o + \frac{(FR_i + FR_o).D.cos.\alpha}{d_m} + \frac{FB}{2}$$
$$F_o^S = \frac{MC_o + MC_i + MER_i + MER_o + MB}{D} + FR_i - \frac{(FR_i + FR_o).D.cos.\alpha}{d_m} - \frac{FB}{2} \qquad [19]$$

where MB and FB are additional braking forces coming from inertia effects or cage effects for example that will be neglected in the current study, and d_m is the pitch diameter. In the latter relationships, the assumption of equal contact angle at inner and outer ring was made which explains why the sliding forces F^S_i and F^S_o are independent of the pivoting moment MP_i and MP_o. More general relations for calculating F^S_i and F^S_o as a function of MP_i, MP_o, α_i and α_o are available, but complex and therefore not shown in this paper whose objective is to given simple relationships and to describe torque concepts mainly.

Similar relationships can be derived for TRB:

$$F_i^S \approx \frac{Mrib + MER_i + MER_o}{D} + FR_o + \frac{(FR_i + FR_o).D.cos\alpha}{d_m} - Frib$$
$$F_o^S = \frac{Mrib + MER_i + MER_o}{D} + FR_i - \frac{(FR_i + FR_o).D.cos\alpha}{d_m} \qquad [20]$$

FINAL TORQUE

Knowing all forces and moments, the final torque around the rotational Y axis due to one rolling element can be derived using the mean inner and outer race radius R_i and R_o (by adding the product Forces. Radius and the moment components in the Y direction).

On the inner and outer race, the torque dT due to one ball is finally:

$$dT = 2.(FR_o + FR_i).\frac{R_o.R_i}{d_m} + \frac{MC_i.R_o + MC_o.R_i}{D} + \frac{MER_i.R_o + MER_o.R_i}{D} + \frac{MP_i.(1 + \frac{D.cos\alpha}{d_m}) + MP_o.(1 - \frac{D.cos\alpha}{d_m})}{2}.sin\alpha \qquad [21]$$

In the latter relationship, the assumption of equal inner and outer ring contact angle is made for keeping the relationship simple. A more general relationship is however available in which different contact angles α_i and α_o (due to centrifugal effects) are considered. Note also that, except for describing the torque due to the pivoting effect, there is no need for using the dissipated power in order to derive the final torque since the forces and moments were calculated as the function of the entrainment and sliding speed and the force equilibrium (in x direction) and moment equilibrium (around the y axis) have been respected.

The moment equilibrium around the z axis has however not been used. Instead, the total power dissipated at the two contacts can be used, leading to:

$$P = MP_i.(\omega_i - \omega_c).sin\alpha + MP_o.\omega_c.sin\alpha \qquad [22]$$

Dividing the latter power by the inner race rotational speed ω_i leads the torque relationship used in Eq. [21].

The final torque can therefore be written:

$$dT = dTR + dTC + dTER + dTP \qquad [23]$$

where dTR, dTC, dTER and dTP are the individual contributions on the torque of the hydrodynamic rolling resistance, curvature effects, elastic rolling resistance, and pivoting effects, respectively.

For TRB the following is obtained:

$$dT = 2.(FR_o + FR_i).\frac{R_o.R_i}{d_m} + \frac{Mrib.R_o}{D} + \frac{MER_i.R_o + MER_o.R_i}{D}. \qquad [24]$$

$$dT = dTR + dTRIB + dTER \qquad [25]$$

where dTRIB is the rib contribution to the torque.

It can be noticed that the component dTR and dTER are analytically identical for TRB and BB.

The final torque is obtained by summation of the individual contribution dT of each rolling element.

ANALYTICAL COMPARISON OF BB *vs.* TRB TORQUE

Having defined analytically how bearing torque is calculated, it is now possible to compare the calculated torque in BB and TRB.

As explained earlier, the hydrodynamic forces FR in BB and TRB are similar if the contact lengths are close, leading to similar hydrodynamic torque if the bearing sizes (race diameter) are similar. BB bearing dimensions are however often larger than TRB ones when a similar contact pressure and bearing life is requested, leading then to a larger BB hydrodynamic torque dTR. There are however cases where a larger contact pressure can be accepted in a BB, leading then to a lower bearing life, but also smaller contact length which can then contribute to a smaller hydrodynamic torque.

Elastic rolling resistance moment dTER in TRB is usually smaller than the BB one because of the smaller TRB roller diameter and rolling element – race friction coefficient. The overall contribution of dTER on the final torque dT is however usually small. (for example 15% of the total torque in a BB).

In the case of a radially loaded BB, the contact angle is nil leading to a maximum value of MC ($f_c = 1$) and a value of MP nil. It can be shown that in this case (and with a/D = 0.25):

$$dTC = 0.16\mu.Q.(\frac{a}{D})^2.d_m \approx 0.01\mu.Q.d_m \qquad [26]$$

The latter torque can be compared to the rib torque (taking f/D = 0.06 , b/f = 2 and ν = 2 degrees) :

$$dTRIB = \mu_{rib}.Q_{rib}.f\sqrt{1 + 0.18(\frac{b}{f})^2}.\frac{R_o}{D} \approx \mu_{rib}.2.sin\nu.Q.f.(1.31).\frac{R_o}{D} \qquad [27]$$

$$dTRIB \approx 0.0027\mu_{rib}.Q.d_m \qquad [28]$$

showing that the rib torque dTRIB can be four times smaller than the curvature torque dTC if the friction coefficients are the same in both bearings.

The roller - rib friction can however be larger than the ball – race one at low speed especially, because the roller – rib surface roughness is often larger than the ball – race one. In this case, the corresponding TRB rib torque (dTRIB) can then be of the same order of magnitude or larger than the BB torque (dTC).

At medium to large speed*viscosity and low rib surface finish, the roller –rib film thickness separates the two surfaces, Λ is larger than 2 and the rib friction coefficient is then of same order of magnitude, or smaller, than the ball – race one, leading to a rib torque dTRIB much smaller than the torque dTC due to the contact curvature in a BB.

Under pure thrust load, the BB contact angle increases leading to a value of MC nil and a maximum value for MP. With a/D = 0.25 the authors obtain:

$$dTP = \frac{3}{8}.\mu.Q.a.sin\alpha \approx 0.09\mu.Q.D \qquad [29]$$

It can be shown that the apex condition in a TRB (roller – race rolling condition) requires:

$$\frac{2.sin\nu}{D} = \frac{sin\alpha}{R_o} \qquad [30]$$

so that the previously described rib torque can be written (with f/D =0.06):

$$dTRIB = \mu_{rib}.Q_{rib}.f\sqrt{1 + 0.18(\frac{b}{f})^2}.\frac{R_o}{D} \approx \mu_{rib}.2.sin\nu.Q.f.(1.31).\frac{R_o}{D} = 1.31\mu_{rib}.Q.f.sin\alpha \qquad [31]$$

$$dTRIB \approx 0.08\mu_{rib}.Q.D \qquad [32]$$

showing again that the TRB rib torque dTRIB and BB pivoting torque dTP are of same magnitude providing the friction coefficient are similar. When the Λ ratio at the rib is larger than 2, the

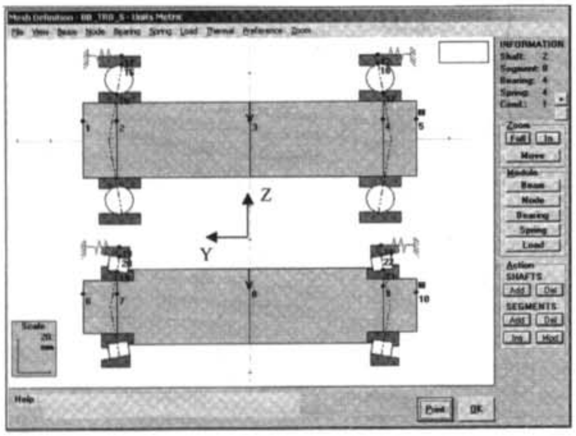

Fig. 9—Sketch of an application.

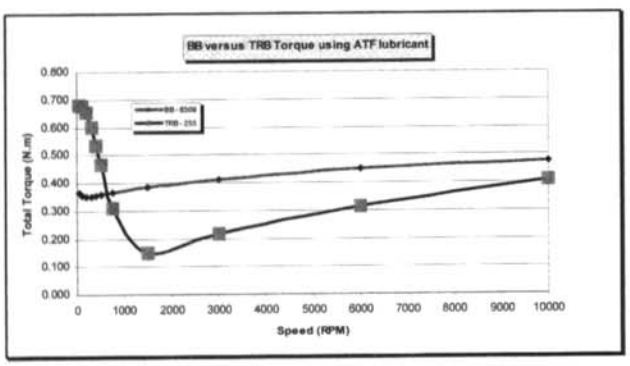

Fig. 10—Numerically calculated ball bearing *vs.* TRB torque in a given case.

rib torque can even be smaller than the pivoting torque because the low roller – rib contact pressure and moderate lubricant shear rate do not always allow the lubricant rib friction coefficient to reach its maximum value (while it is easily reached at the ball – race contact).

NUMERICAL COMPARISON BB *vs.* TRB TORQUE

A numerical tool called SYSx has been developed by Houpert and Hauswald and is described in (Houpert, (1995), (1997), Houpert, et al., (1998), Hauswald, et al., (2000)).

This tool enables us to include the shaft and housing elasticity as well as the gear loading in the calculations of bearing performance (stiffness, life, torque, etc.). Advanced features such as manufacturing tolerances, profile, centrifugal and roller skewing effects on torque are also included, (although not described in this paper which only describes, for the sake of simplicity, some nominal or basic torque concepts).

All bearing types can be calculated and Fig. 9 shows a simple sketch of the case analyzed in the next example.

A load of 25000 N is applied on a 45 mm bore shaft supported by two 6309 ball bearings having a pitch diameter of 72.5 mm,

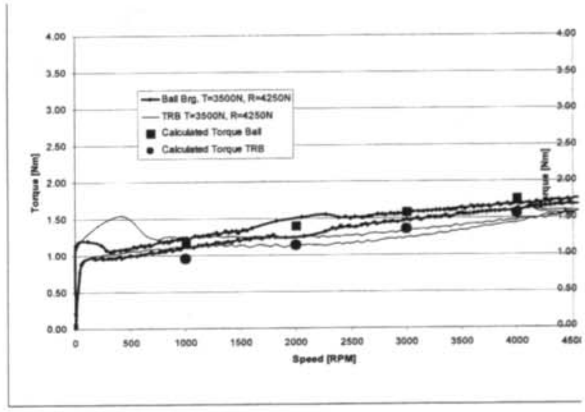

Fig. 11—Torque measurements and calculations @ 50°C – comparison TRBs *vs.* floating ball bearing, taken from (Gradu, (2000)).

8 balls of diameter 17.462 mm, a K race factor of 0.0354 (i.e. an osculation ratio f of 0.5177), an initial contact angle of 10.035 degrees and a composite ball – race RMS surface finish of 0.035 micrometer. On the most loaded ball, the calculated load is 6970 N, the Hertzian pressure is 2770 MPa and the contact ellipse length is 6.8 mm.

An ATF lubricant having a viscosity of 26 and 5.5 cSt at 40° and 100°C respectively is used with an operating temperature of 60°C. Its relatively low viscosity is suitable for reducing the hydrodynamic rolling force at large speed, but is not appropriate for reducing asperity effects on the friction coefficient, hence rib torque for example at low speed.

The C1 capacity of the ball bearing is 52000 N, leading to an estimated C90 capacity of 11600 N.

A slightly smaller TRB (255/253) of similar capacity was found, C90 capacity of the TRB being 13400 N. This bearing has 14 rollers of mean diameter 8.9612 mm, effective roller length of 10.7 mm, a cup contact angle of 10.33 degrees and a composite RMS surface finish of 0.11 and 0.09 micrometer at the roller - race and roller-rib contact, respectively. The roller – rib contact geometry leads to a contact height f of 0.92 mm. On the most loaded roller, the load is 3831 N, the contact pressure at the center of the contact is 1999 MPa while the calculated edge stress on the cone is 3453 MPa.

Note that it was not possible to find in this case a TRB with a contact length equal to the BB one. The effective roller length of the roller (10.7 mm) is larger than the BB contact length (6.8 mm at the tested load). This will therefore cause hydrodynamic forces in the TRB to be slightly larger. The number of rolling elements in the TRB being also larger than the one in the BB, a larger final hydrodynamic TRB torque can be expected.

Despite this fact, it can be seen on the Fig. 10 that the final TRB torque calculated *vs.* speed is smaller than the BB one at medium and large speed. This is due partly due to the fact that the TRB mean race radii (cone and cup) are smaller than the BB ones. At low speed, the roller-rib Λ ratio becomes small causing the friction coefficient and final rib torque to be larger than the BB race torque due to pivoting.

The calculated TRB lief is 1.8 times larger than the BB one.

Choosing a smaller TRB would result in a lower TRB life, but than also smaller TRB torque.

Many SYSx additional graphical outputs are available as for example the calculated shaft and bearing deflections. Shaft deflections are smaller when TRB are used because of the larger TRB bearing stiffness.

EXPERIMENTAL COMPARISON BB *vs.* TRB TORQUE

Such favorable torque results for TRB have been confirmed experimentally by M. Gradu in (Gradu, (2000)) where a BB and TRB used in an automotive transmission have been calculated using SYSx (hence the same herein described torque model) and tested *vs.* speed at several loads.

As an example, Fig. 11 taken from (Gradu, (2000)), shows the torque measured and calculated at 50°C, with a radial and axial bearing load of 4250 and 3500 N, respectively. The floating BB is a 6305 and is compared to the TRB (07100/07196). The lubricant used is ATF (Automatic Transmission Fluid). More details about the bearing tested and operating conditions can be found in (Gradu, (2000)).

Results confirm experimentally the lower TRB torque calculated in this application, (partly due to the smaller TRB size), and also show the good correlation obtained when comparing calculated and measured torques.

CONCLUSIONS

BB and TRB torque models have been analytically described for outlining common points and differences.

BB and TRB bearing torque- due to hydrodynamic losses- can be similar, as long as the rolling element – race contact lengths are similar or the TRB size is smaller than the BB one. The latter condition can be found when large ball diameter (larger than the roller diameter) is used for decreasing the Hertzian pressure and competing with the TRB life. The TRB roller length is however often larger that the Ball – race contact length, but this detrimental effect on the hydrodynamic torque is also often compensated by the smaller size of the TRB, leading finally to similar hydrodynamic torque.

The sliding speed encountered at the ball – race contact has been described and is responsible for two BB torque contributions: the torque due to the curvature in space of the contact ellipse and the torque due to spinning effects.

These latter torque contributions don't exist in TRB. But TRB are subjected to a roller – rib torque, found especially at low speed or low roller – rib film thickness.

Analytical relationships have been derived allowing a direct comparison between TRB and BB torque as a function of the operating conditions and bearing geometry.

Providing the roller – rib friction coefficient can be held low, (condition occurring when the roller - rib film thickness is larger than the composite surface finish, i.e. at medium to large speed*viscosity with low roller – rib surface finish), the TRB torque can be smaller than BB torque which has been demonstrated experimentally in reported literature.

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